Multi-Agent AI System for Cognitive Digital Twins in Education

Graph-based subject modeling · learner digital twin · predictive adaptive tutoring

PROJECT PURPOSE

Build a multi-agent learning system that structures a subject into a knowledge graph, estimates the learner's current understanding with a **cognitive digital twin**, and continuously adapts instruction using predicted versus observed learning behavior.

THREE DISTINCT SPACES

The twin is an estimate, not the subject itself

E

SUBJECT / TARGET GRAPH

Complete knowledge space for the scoped topic. Nodes represent concepts; subgraphs, subconcepts; edges, weighted and annotated relations; layers and states capture different views.

F

ACTUAL UNDERSTANDING

The learner's real, changing understanding. It cannot be inspected directly: the system only sees answers, questions, hypotheses, and observations.

F′

COGNITIVE DIGITAL TWIN

A learner-specific graph estimating F against the structure of E. Initialized by profiling and refined whenever observed responses diverge from predictions.

INITIAL SETUP

Before the repeated learning cycle

1. Select & scope an existing or new subject → **2. Research agent** collects web sources and constructs E → **3. Profile** the learner using diagnostic questions to initialize F′.

CONTINUOUS PREDICT–TEACH–EVALUATE–UPDATE LOOP

Repeated after every learning activity

01 PLAN

Choose next lesson using E and F′

→

02 PREDICT

Estimate expected learner response

→

03 TEACH + OBSERVE

Teach, quiz; gather actual output O

→

04 COMPARE + UPDATE

Calculate mismatch; correct F′

REPEAT: select the next lesson with the revised twin

CONCEPTUAL MATHEMATICAL FORMULATION

No specific metric or algorithm is assumed

$$L_{t+1} = \pi(E, F'_t) \qquad \widehat{O}_{t+1} = P(F'_t, L_{t+1})$$

$$\varepsilon_{t+1} = d(\widehat{O}_{t+1}, O_{t+1}) \qquad F'_{t+1} = U(F'_t, O_{t+1}, \varepsilon_{t+1})$$

LEARNING OBJECTIVE — REDUCE THE KNOWLEDGE GAP

$$\min_{\text{adaptive lessons}} D(F'_t, E)$$

L = selected lesson; Ô = predicted learner output; O = actual answers, questions, hypotheses and observations. d measures prediction mismatch; U uses that evidence to update F′.

Key distinction: adaptive teaching aims to reduce the gap between F′ and E **through real learning**; the prediction error keeps F′ calibrated to F. Updating the twin alone does not establish learning.

REQUIRED TECHNOLOGIES & FRAMEWORKS

Spring · JPA · Servlet · REST · gRPC · Asynchronous communication · Logback · Unit tests · Integration tests · Circuit breaking & retries · Kong · Eclipse Ditto · Vert.x · NARU · Pyramid Architecture · Nuts